AI Lawyer

A grounded AI legal helper for everyday criminal-law problems in India

SOLUTION DOCUMENT

Bharatiya Nyaya Sanhita (BNS), 2023

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1. The problem

Most people who are threatened, cheated, robbed or harassed do not know which law applies, what proof matters, or how to approach the police. Lawyers are expensive, and general-purpose chatbots can **invent law**: wrong section numbers, made-up punishments and fake case citations. For a legal topic, a confident wrong answer is worse than no answer.

2. The solution

AI Lawyer is a conversational helper that turns a messy story, told in English, Hindi or Hinglish, into a structured case and explains where the person stands using **only verified text of the Bharatiya Nyaya Sanhita (BNS)**. Every answer is checked in code before it is shown.

- **Case builder:** what happened, when, who, what proof exists, harm suffered and what is still missing, shown live in a sidebar.
- **Plain-language guidance:** the BNS sections that may apply, practical options and next steps.
- **Document drafting:** police complaint, demand/legal notice or a one-page summary for a lawyer, with unknown details left as [PLACEHOLDERS].
- **Practice mode:** the AI plays the other side (e.g. the police officer) so the user can rehearse, then gives concrete feedback.
- **Safety routing:** emergencies show helplines (112, 181, 1098, 1930); people accused or served a notice are directed to a real lawyer.

3. Architecture

The assistant is a **LangGraph** state machine. Each message is first triaged, then routed to the right node. Dotted lines in the diagram are conditional routes.

triage classifies intent (discuss, draft, practice), domain (criminal, civil, unclear), emergency and accused/notice flags, and writes an English search query.
emergency / handoff end the turn with helplines or a lawyer referral, with no DIY advice.
update_case merges new facts into the structured case file without inventing anything.
retrieve looks up sections the user named, runs semantic search, applies a distance gate, has an LLM judge fit, and builds the allow-list.
advise / draft write the reply or document using only the allowed sections as sources.
verify scans the reply for section numbers and case citations; one rewrite, then redaction.
practice / practice_feedback run the role-play and the closing feedback.

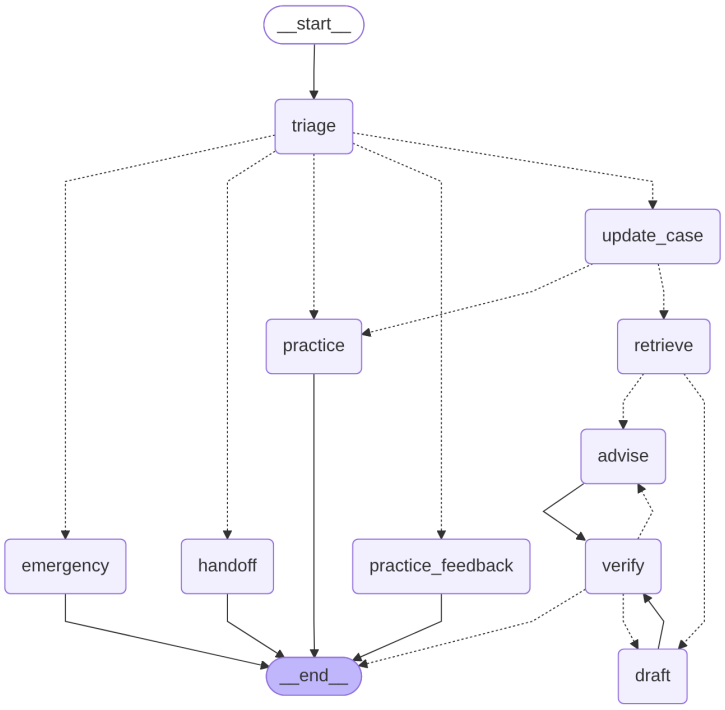


Figure: LangGraph workflow of the AI Lawyer chatbot (generated from the compiled graph).

4. How the solution avoids invented law

Hallucination is handled in layers, and the final layers are deterministic code, not the LLM.

Layer	What it does
Distance gate	If the best retrieved section is too far from the query (cosine distance above 0.50), no section is offered at all.
LLM fit judge	Each candidate is rated strong, possible or no fit against the facts; only strong and possible survive.
Allow-list	The writer may cite only sections that passed the judge, plus numbers the user typed.
Verify node	A regex extracts every section number in the reply. Unknown numbers trigger one rewrite, then are redacted. Case-law citations are blocked.
Exact lookup	If the user names a section, it is checked against the index; a non-existent one is reported as missing. IPC numbers are never mapped from memory.
Scope honesty	Civil matters (deposit, salary, refund) get case-building help but no statute citation.
Escalation	Emergency, accused or notice-received cases are routed away from DIY mode.

5. Knowledge base and retrieval

- **Data:** 358 BNS sections (chapter, section name, full text) from a public Kaggle dataset.
- **Embedding:** BAAI/bge-m3, a multilingual model, so Hindi and Hinglish queries match English statute text. One record per section, no chunking needed.
- **Index:** ChromaDB with cosine similarity, built automatically on first run and stored locally.
- **Search:** the plain-English query and the raw user message are both searched and merged (top 5).

6. Technology stack

Layer	Choice	Layer	Choice
UI	Streamlit chat app	**Vector store**	ChromaDB
Orchestration	LangGraph + LangChain	**Embeddings**	BAAI/bge-m3 (FlagEmbedding)
LLM	Google Gemini (2.5 Flash-Lite default)	**Memory**	LangGraph MemorySaver (per session)

7. Example flow

User: "A person took Rs. 50,000 from me on a false promise and vanished." **System:** triage marks it criminal, update_case records amount and other party, retrieve finds cheating-related BNS sections and the judge keeps those that fit, advise explains them in plain words with options (preserve bank proof, write the facts down, approach the police station) and asks up to three follow-up questions, verify confirms every cited number is on the allow-list. The user can then ask for a draft police complaint or practise the conversation with the police officer.

8. Limitations and next steps

- Covers criminal law under the BNS only; no BNSS, IT Act, other statutes, case law or IPC-to-BNS mapping.
- Conversation memory is in-process and lost on restart; LLM output can still be incomplete, so a lawyer should always confirm anything serious.
- **Next:** IPC-to-BNS mapping, more offence categories and languages, persistent storage, and an evaluation set to tune the 0.50 distance threshold.

Disclaimer: AI Lawyer is not a lawyer and does not give legal advice. Free legal aid: NALSA helpline 15100. Emergency: 112.